

Question Paper

Exam Date & Time: 12-Jun-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. DEGREE EXAMINATIONS - JUNE 2025

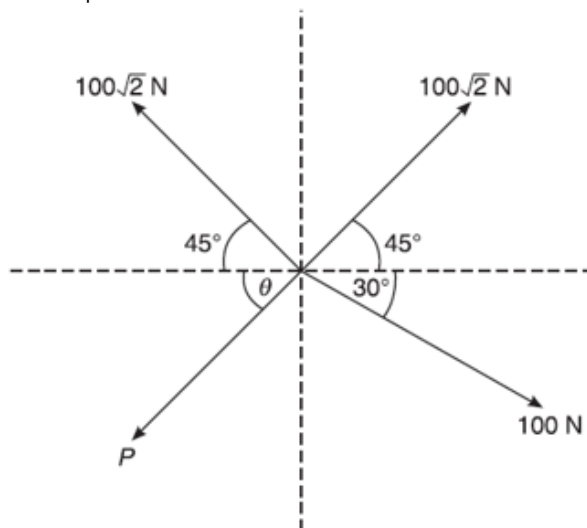
SUBJECT: CIE 1071/CIE_1071 - MECHANICS OF SOLIDS
(MAKE-UP)

Marks: 50

Duration: 180 mins.

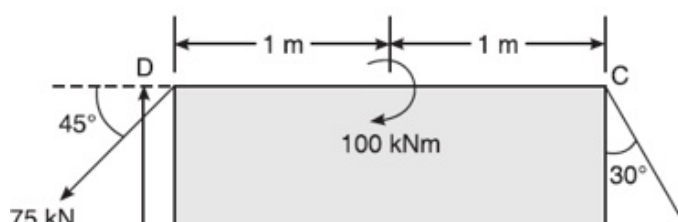
Answer all the questions.

- 1A) For the figure shown below, determine the magnitude and direction of force P , which keeps the concurrent force system in equilibrium.. (5)



Course Outcome: "CO2" / Cognitive Level: "Application"

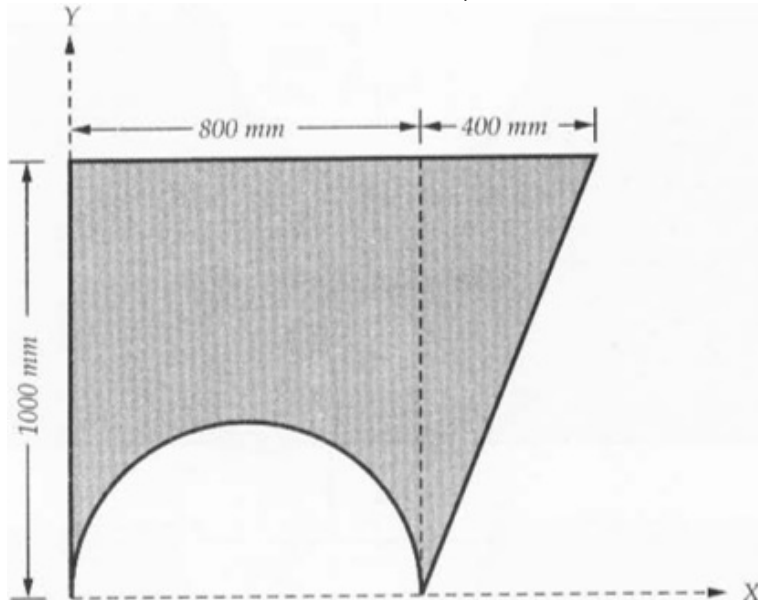
- 1B) Determine the magnitude, direction and distance of the resultant from point 'A' for the system of forces shown in Figure. (5)





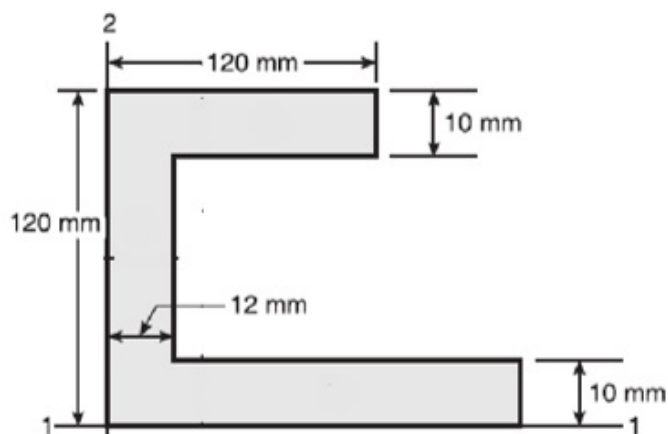
Course Outcome: "CO1" / Cognitive Level: "Understand"

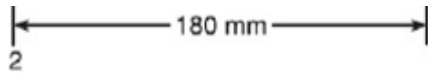
- 2A) Locate the centroid of the shaded area with respect to the X- axes shown in the figure. (3)



Course Outcome: "CO3" / Cognitive Level: "Application"

- 2B) Determine the moment of inertia of the section about 1-1 and 2-2 axes, as shown in the figure. (5)





Course Outcome: "CO3" / Cognitive Level: "Application"

- 2C) Derive the centroid of the triangular lamina from the principle of moments with respect to the base of the triangle. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3A) A cylindrical vessel 2m long has an internal diameter of 300mm and wall thickness of 10mm. If it subjected to internal pressure of 90 m of water. Calculate the hoop stress, longitudinal stress, change in diameter, length and volume. (5)
Take $E = 150 \text{ GPa}$ and $\mu = 0.28$.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 3B) Show that when a tapered bar of length L and thickness b , having rectangular cross section is subjected to an axial tensile load P , it undergoes total extension equal to (5)

$$\frac{2.302 \times P \times L}{b \times E \times (d_2 - d_1)} (\log d_2 - \log d_1)$$



Course Outcome: "CO4" / Cognitive Level: "Understand"

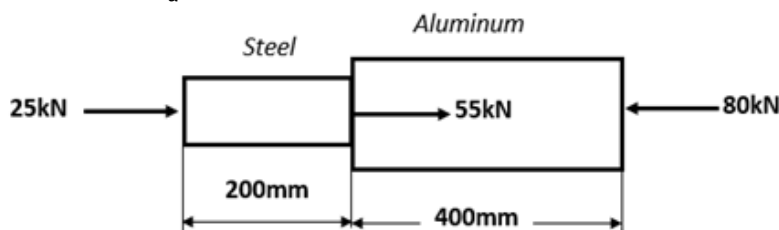
- 4A) An aluminum rod of 20 mm diameter and 60 mm length is subjected to a compressive force of 250 kN. If modulus of elasticity and modulus of rigidity of aluminum is 70 GPa and 27 GPa respectively, determine bulk modulus, Poisson's ratio, change in length and change in diameter of the bar. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4B) Distinguish between the behavior (stress vs strain graph) of mild steel rod and caste iron rod when subjected to uniaxial tensile load. (3)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 4C) A composite bar consists of a steel rod (20 mm diameter) rigidly fastened to an aluminium rod (60 mm diameter) as shown in figure. Axial loads are applied at the positions indicated. Determine the stress and strain in both rods. $E_s = 200 \text{ GPa}$ and $E_a = 70 \text{ GPa}$. (3)

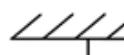
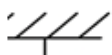


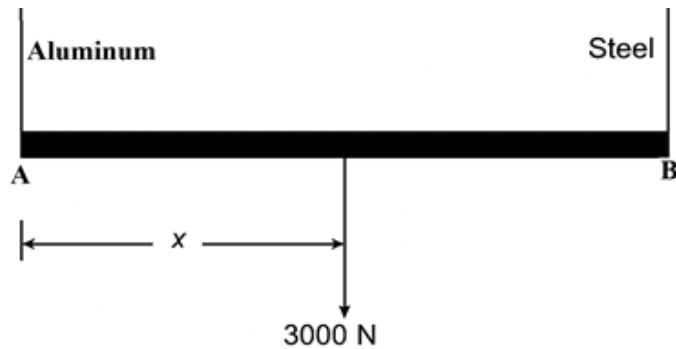
Course Outcome: "CO4" / Cognitive Level: "Application"

- 5A) A rigid bar AB is 10 m long and is suspended by two vertical rods at its ends A and B. The bar remains horizontal under initial conditions. (5)

Sl No.	Material	Length (m)	Cross-sectional area (mm^2)	Modulus of Elasticity (GPa)
1	Steel	2.5	500	200
2	Aluminum	2.5	700	70

A load $P=3000 \text{ N}$ is applied at a distance x from point A. Determine the position x such that the bar remains horizontal.





Course Outcome: "CO5" / Cognitive Level: "Application"

- 5B) A composite bar is made of three segments-Steel, Brass, and Aluminum connected in series. The bar is rigidly fixed at both ends and a temperature rise of 60°C occurs. (5)

Given:

Material	Length (mm)	Area (mm^2)	Young's Modulus (GPa)	Coefficient of thermal expansion (α , $^{\circ}\text{C}$)
Steel	150	300	200	12×10^{-6}
Brass	200	400	100	19×10^{-6}
Aluminum	250	500	70	23×10^{-6}

Determine the thermal stress developed in each material assuming supports are rigid.



Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025

CIE 1071-PHY:MECHANICS OF SOLIDS

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	5
3	3	CO 3	10
4	4	CO 4	20
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name	Subject
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MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025

CIE 1071-PHY:MECHANICS OF SOLIDS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	13(26%)
3	Application	9	37(74%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		12	50 (100%)